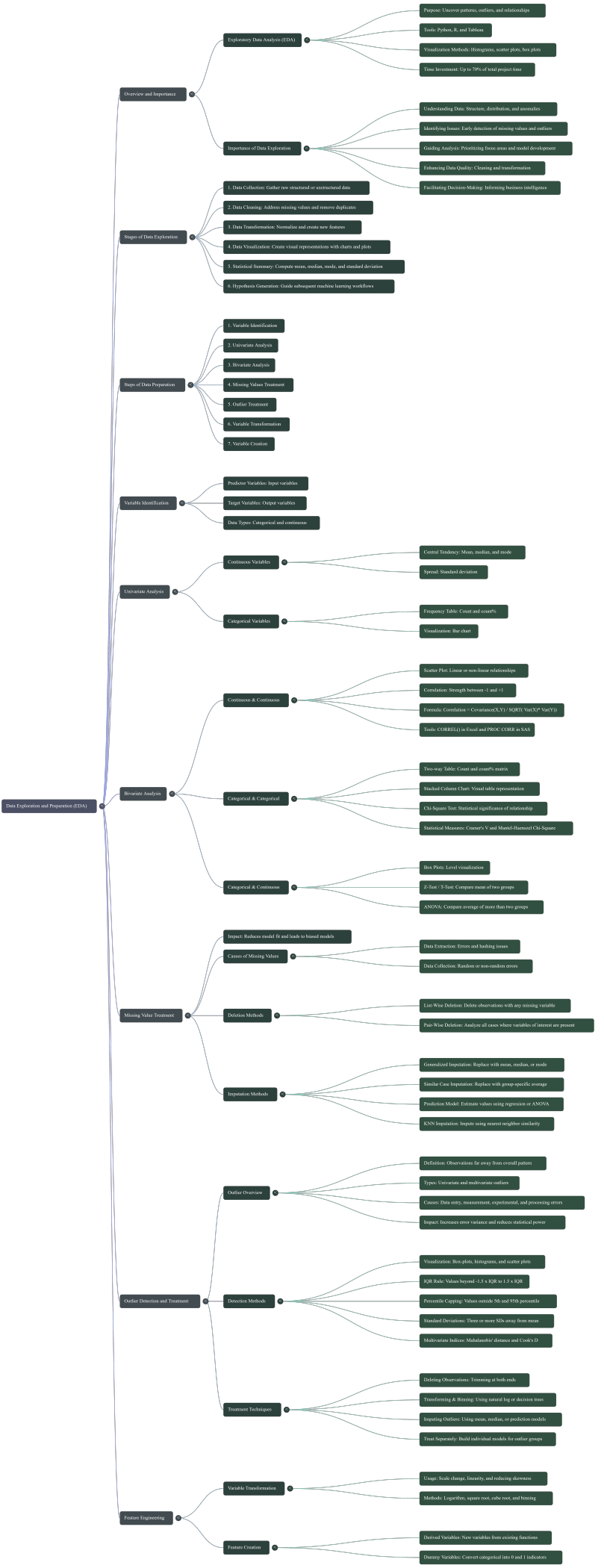

Data Exploration and Preparation (EDA)
Overview and Importance
Exploratory Data Analysis (EDA)
Purpose: Uncover patterns, outliers, and relationships
Tools: Python, R, and Tableau
Visualization Methods: Histograms, scatter plots, box plots
Time Investment: Up to 70% of total project time
Importance of Data Exploration
Understanding Data: Structure, distribution, and anomalies
Identifying Issues: Early detection of missing values and outliers
Guiding Analysis: Prioritizing focus areas and model development
Enhancing Data Quality: Cleaning and transformation
Facilitating Decision-Making: Informing business intelligence
Stages of Data Exploration
1. Data Collection: Gather raw structured or unstructured data
2. Data Cleaning: Address missing values and remove duplicates
3. Data Transformation: Normalize and create new features
4. Data Visualization: Create visual representations with charts and plots
5. Statistical Summary: Compute mean, median, mode, and standard deviation
6. Hypothesis Generation: Guide subsequent machine learning workflows
Steps of Data Preparation
1. Variable Identification
2. Univariate Analysis
3. Bivariate Analysis
4. Missing Values Treatment
5. Outlier Treatment
6. Variable Transformation
7. Variable Creation
Variable Identification
Predictor Variables: Input variables
Target Variables: Output variables
Data Types: Categorical and continuous
Univariate Analysis
Continuous Variables
Central Tendency: Mean, median, and mode
Spread: Standard deviation
Categorical Variables
Frequency Table: Count and count%
Visualization: Bar chart
Bivariate Analysis
Continuous & Continuous
Scatter Plot: Linear or non-linear relationships
Correlation: Strength between -1 and +1
Formula: Correlation = Covariance(X,Y) / SQRT(Var(X)* Var(Y))
Tools: CORREL() in Excel and PROC CORR in SAS
Categorical & Categorical
Two-way Table: Count and count% matrix
Stacked Column Chart: Visual table representation
Chi-Square Test: Statistical significance of relationship
Statistical Measures: Cramer's V and Mantel-Haenszel Chi-Square
Categorical & Continuous
Box Plots: Level visualization
Z-Test / T-Test: Compare mean of two groups
ANOVA: Compare average of more than two groups
Missing Value Treatment
Impact: Reduces model fit and leads to biased models
Causes of Missing Values
Data Extraction: Errors and hashing issues
Data Collection: Random or non-random errors
Deletion Methods
List-Wise Deletion: Delete observations with any missing variable
Pair-Wise Deletion: Analyze all cases where variables of interest are present
Imputation Methods
Generalized Imputation: Replace with mean, median, or mode
Similar Case Imputation: Replace with group-specific average
Prediction Model: Estimate values using regression or ANOVA
KNN Imputation: Impute using nearest neighbor similarity
Outlier Detection and Treatment
Outlier Overview
Definition: Observations far away from overall pattern
Types: Univariate and multivariate outliers
Causes: Data entry, measurement, experimental, and processing errors
Impact: Increases error variance and reduces statistical power
Detection Methods
Visualization: Box-plots, histograms, and scatter plots
IQR Rule: Values beyond -1.5 x IQR to 1.5 x IQR
Percentile Capping: Values outside 5th and 95th percentile
Standard Deviations: Three or more SDs away from mean
Multivariate Indices: Mahalanobis' distance and Cook's D
Treatment Techniques
Deleting Observations: Trimming at both ends
Transforming & Binning: Using natural log or decision trees
Imputing Outliers: Using mean, median, or prediction models
Treat Separately: Build individual models for outlier groups
Feature Engineering
Variable Transformation
Usage: Scale change, linearity, and reducing skewness
Methods: Logarithm, square root, cube root, and binning
Feature Creation
Derived Variables: New variables from existing functions
Dummy Variables: Convert categorical into 0 and 1 indicators